

Introduction to Git Version Control

CMPUT 301

Version Control

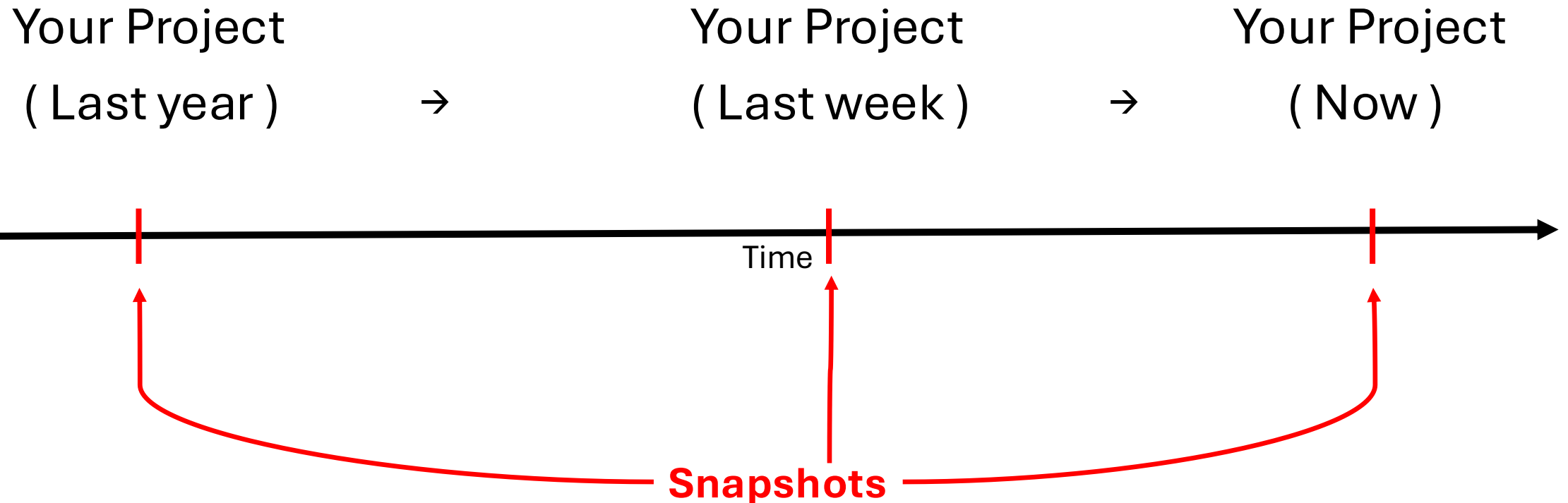
- Your Project changes over time:
 - Today: it runs
 - Yesterday: it mostly works
 - Last week: it did not fully work
 - Last year: it may have been something totally different

Version Control

- Version control tracks:
 - Changes
 - Files
 - Authors
 - Time

Version Control

- Version control snapshots a state of the project at a point in time

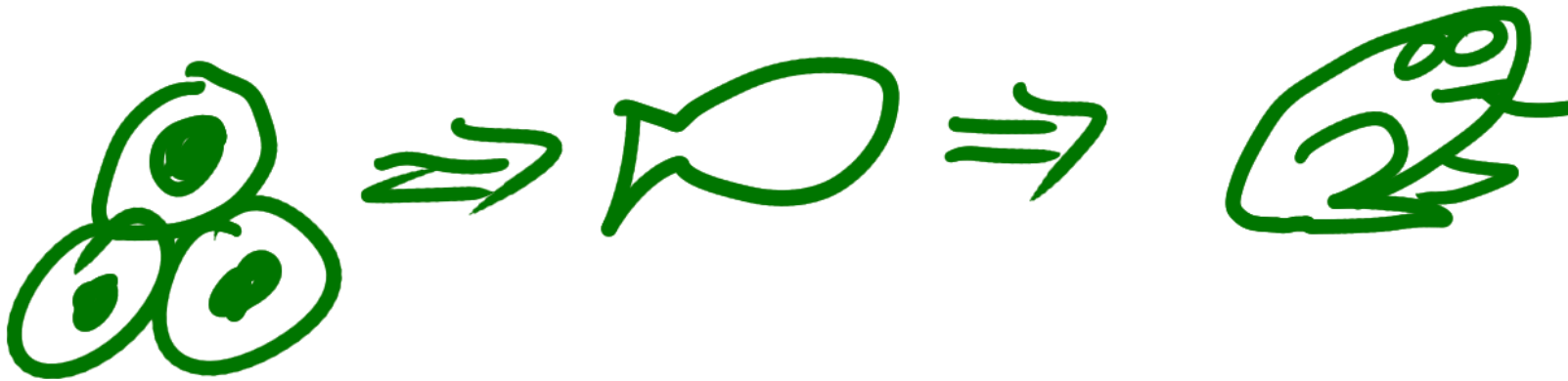


Software Has Context

- A project is not just code
- A project has:
 - Current time
 - Current requirements
 - Current functionality
 - Current issues
 - Current team

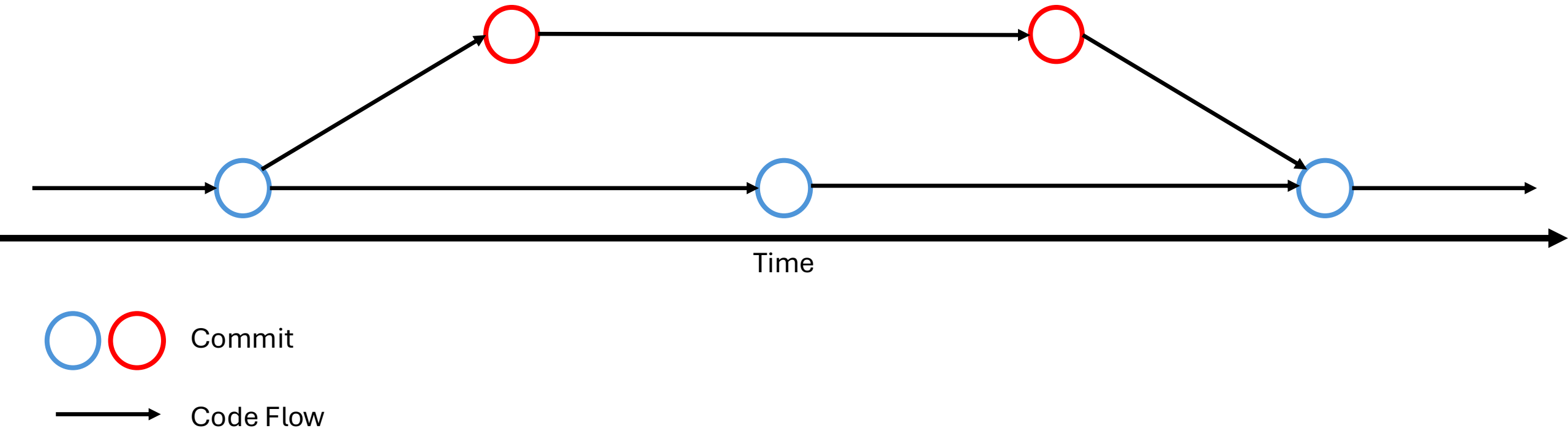
Software Evolves

- Version control helps track evolution



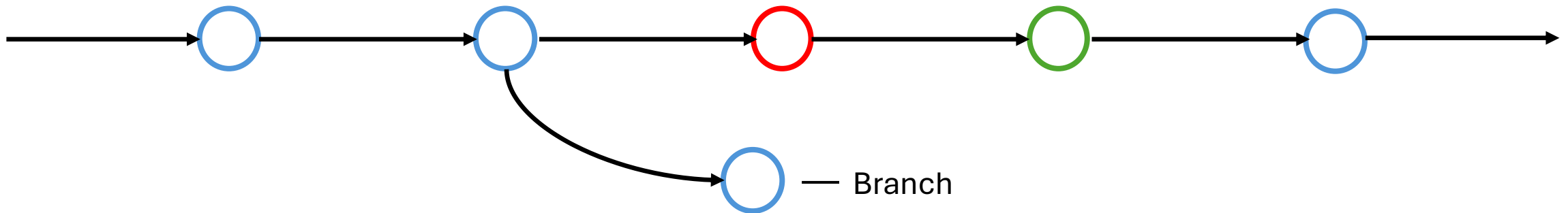
Commits Record Change

- Each commit records a meaningful change to the project



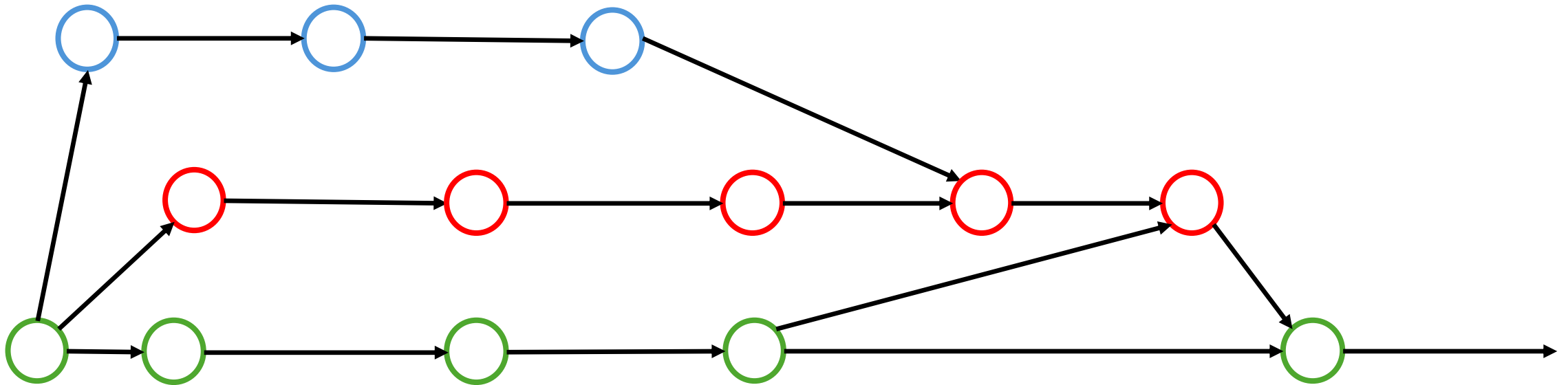
Centralized Version Control

- A centralized system has one main repository that users interact with



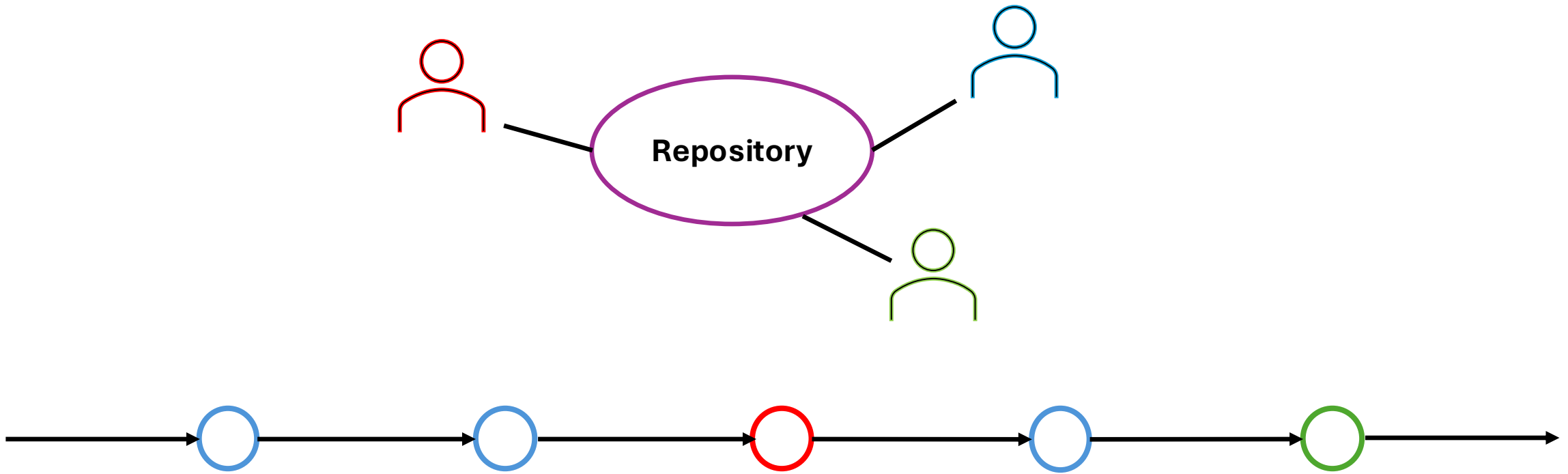
Distributed Version Control

- Each repository is like a branch
- (Repo, Branch) tuple is a unique branch
- Each developer has a full repository with its own history

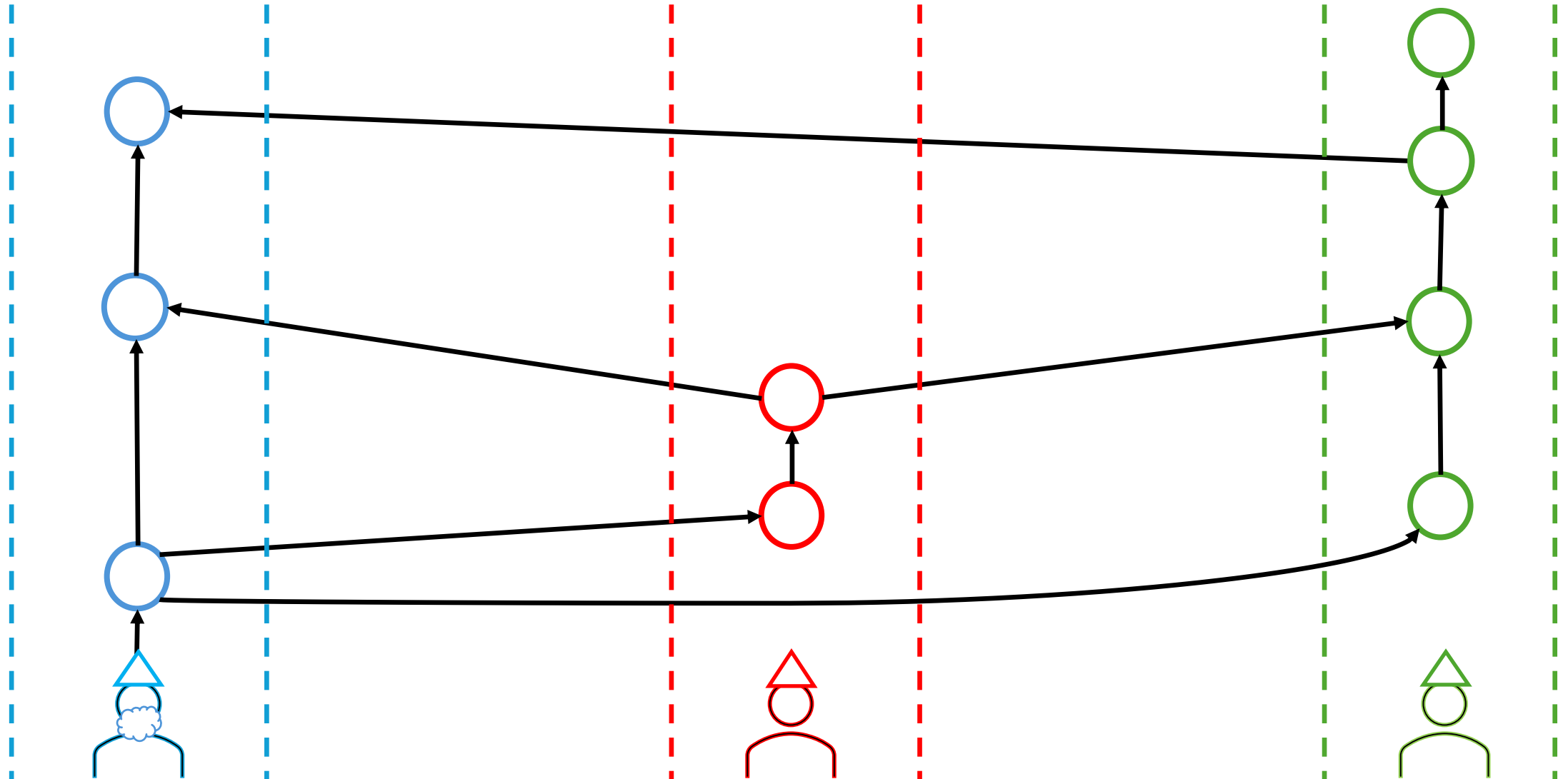


Repository Style: Central Repository

- Shared place for updates, checkouts, modifies, and commits



Repository Style: Separate Towers



Repository Style: Separate Towers

- Only need to give READ access
- No need to give write access
- Provide isolation from change
- Each repository is a personal branch